Railway Reservation System using C++

Console-based reservation system managing seat booking and cancellation through file storage

Problem Statement

Challenge

Ticket booking requires accurate seat management to prevent duplicate bookings and maintain correct seat status.

Manual System Limitations

- Duplicate bookings possible
- Incorrect seat status records
- Need for computerized solution

My Contribution to the Project

Seat Booking Logic

Designed and implemented core booking functionality with availability checks

Cancellation Mechanism

Implemented seat release system with proper status updates

Business Rules

Applied validation logic and constraints to ensure data integrity

Booking and Cancellation Workflow

01

Availability Check

Booking allowed only for available seats

02

Double Booking Prevention

System prevents duplicate reservations

03

Cancellation Validation

Only booked seats can be cancelled

04

Immediate Update

Seat status updated in real-time

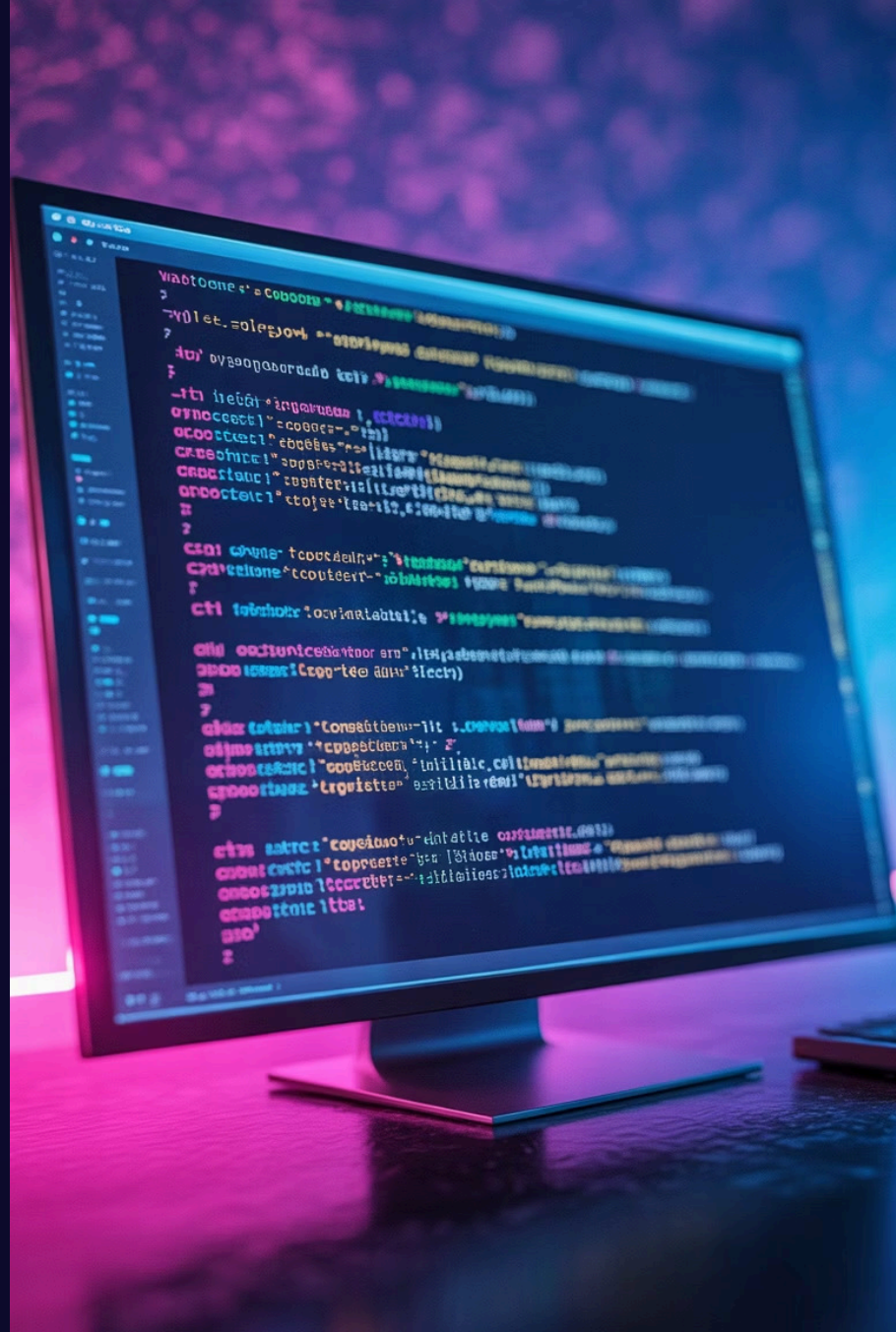
Validation and Business Rules

Input Validation

- Valid seat range: 1–50
- Booking blocked for reserved seats

Error Handling

- Cancellation blocked for empty seats
- All invalid actions handled safely



C++ Concepts and Conclusion

Arrays

Used for seat management and status tracking

File Handling

Implemented data persistence through file storage

Exception Handling

Applied error control mechanisms

Conclusion: System ensures reliable booking and cancellation through practical application of C++ programming concepts.